Does post-treatment MRI add to clinical information for short-term progression prediction in glioblastoma?

A censoring-aware re-evaluation on MU-Glioma-Post with external testing on LUMIERE

PRISM project — internal technical report

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Abstract

Aim. To test whether information from a single post-treatment MRI improves prediction of progression within 120 days over information that is available without imaging (clinical, timing and treatment). **Data.** MU-Glioma-Post (University of Missouri; 314 eligible surveillance scans from 157 patients, of which 253 scans / 128 patients are WHO grade 4) for development, and LUMIERE (132 scans / 61 glioblastoma patients) for external testing. **Methods.** Censoring-aware labels; 5×5 repeated patient-grouped cross-validation; IPCW AUC at 120 days with patient-cluster bootstrap confidence intervals; nested selection over feature groups and over 70 radiomics configurations; paired ΔAUC against a non-imaging baseline. **Results.** Demographics and molecular markers alone were weak (AUC 0.56–0.63, grade 4). Adding timing raised the AUC to 0.758 (gradient boosting). Imaging-only models reached at most 0.660. Adding imaging to the non-imaging model did not improve it reliably; in the honest nested estimate, radiomics changed the AUC by -0.008 (grade 4) and $+0.013$ (all gliomas), and none of 70 radiomics configurations had a ΔAUC confidence interval above zero. On the PRISM report split, a model without imaging scored 0.870–0.901 against 0.802 for the PRISM report radiomics model. On LUMIERE, the PRISM report radiomics model dropped to 0.61, while a clinical + timing + volume model reached 0.72. **Conclusion.** In this setting (conventional MRI, one development centre, short-horizon progression), imaging adds no measurable value over clinical and treatment information.

1 Background

Surveillance MRI after treatment of glioblastoma is used to decide when to intervene, and several groups have proposed image-based models that predict progression from a follow-up scan. Reported discrimination on MU-Glioma-Post is around 0.80, but these figures come from single train/test splits and are rarely compared with a baseline that uses the clinical and treatment information already available on the day of the scan. The PRISM report radiomics model (AUC 0.802 on a 30-patient test split) is one such model.

This report asks a narrow question: *once clinical, timing and treatment information is in the model, does the image add anything?*

2 Data and endpoint

2.1 Development cohort: MU-Glioma-Post

The TCIA MU-Glioma-Post collection contains 596 post-treatment timepoints from 203 patients, with T1, T1c, T2 and FLAIR images, expert-refined four-label masks (enhancing tumour, non-enhancing core, surrounding FLAIR hyperintensity, resection cavity) and a clinical workbook. Eligibility rules were applied in order (Table 1). Index scans are surveillance scans taken before first progression and before any late treatment (second-line therapy, immunotherapy, or new treatment).

Table 1: Cohort flow (MU-Glioma-Post).

Step	Scans	Patients	Excluded
All timepoints	596	203	–
All four sequences and mask present	594	203	2
Tumour (labels 1–3) present	592	203	2
MRI day known	589	202	3
Before late treatment	508	199	81
Before first progression (index scans)	**314**	**157**	194
of which WHO grade 4	253	128	

2.2 Endpoint

For each index scan, the label at horizon $H = 120$ days is 1 if progression is documented within H days, 0 if progression occurs later, the patient died without progression, or progression-free follow-up reaches H days, and censored otherwise. Follow-up uses the latest dated MRI or clinical event. At 120 days the cohort has 99 positive, 178 negative and 37 censored scans (grade 4: 92 / 136 / 25).

2.3 External cohort: LUMIERE

LUMIERE (Suter et al., 2022) provides expert RANO ratings per timepoint for 91 glioblastoma patients. Index scans are rated follow-ups (stable disease, partial or complete response, post-operative; at least two weeks after surgery) before the first RANO progression. With the same 120-day rule this gives 132 scans from 61 patients (47 positive, 77 negative, 8 censored). LUMIERE has no treatment fields, so only features available in both cohorts (age, sex, IDH/MGMT, time since surgery, early post-radiotherapy flag, compartment volumes) are used externally.

3 Methods

3.1 Feature groups

- **Clinical**: age, sex, IDH and MGMT status with an explicit "unknown" level.
- **Timing**: days since diagnosis, days since end of radiotherapy, early post-radiotherapy flag (within 12 weeks), visit index.
- **Treatment**: radiotherapy given, dose, fractions, hypofractionation, start day and duration; initial chemotherapy; day of surgery; biopsy before resection; previous brain tumour.
- **Volumes**: log volumes of each mask compartment.
- **Morphology**: enhancing fraction, multifocality, largest axial enhancing area, distance of enhancement to the cavity, intensities relative to normal-appearing brain.
- **Change**: differences from the previous scan and from the nadir, growth rates, a $\geq 40\%$ enhancing-volume increase flag, new enhancing component.
- **Radiomics**: the PRISM report features (two grey levels) and corrected PyRadiomics features (whole-brain normalisation, 32 bins, per compartment), with 70 configurations of stability filtering, feature selection and integration (concatenation or a stacked radiomics score).

3.2 Models and evaluation

Two learners were used: ridge logistic regression with the penalty chosen by inner patient-grouped cross-validation, and shallow gradient boosting. All preprocessing and selection were fitted within training folds. Models were evaluated with 5×5 repeated patient-grouped stratified

cross-validation, using inverse-probability-of-censoring-weighted (IPCW) cumulative/dynamic AUC at 120 days. Confidence intervals come from a patient-cluster bootstrap (1000 resamples). Paired ΔAUC was computed against the ridge non-imaging baseline. Nested selection over feature groups and over radiomics configurations gives an estimate that includes the cost of choosing a configuration. A label-permutation control returned AUC ≈ 0.50.

4 Results

4.1 Imaging versus non-imaging information

Table 2 compares models by the information they use.

Table 2: IPCW AUC at 120 days, MU-Glioma-Post grade 4 (253 scans, 128 patients), repeated patient-grouped cross-validation, 95% patient-cluster bootstrap CI.

Information used	Ridge	Boosting
Without imaging		
Clinical only (age, sex, IDH/MGMT)	0.556 [0.456, 0.647]	0.634 [0.545, 0.713]
Clinical + timing	0.662 [0.583, 0.754]	**0.758** [0.677, 0.838]
Clinical + timing + treatment	0.694 [0.613, 0.775]	0.723 [0.633, 0.806][a]
Imaging only		
Volumes + morphology + change	0.603 [0.515, 0.686]	0.545 [0.462, 0.628]
+ PRISM report radiomics	0.660 [0.589, 0.736]	0.652 [0.567, 0.730]
Non-imaging + imaging		
Clinical + timing + volumes	0.681 [0.604, 0.768]	0.733 [0.660, 0.805]
Clinical + timing + radiomics	0.709 [0.631, 0.783]	0.740 [0.669, 0.816]
Clinical + timing + all imaging	0.687 [0.610, 0.767]	0.717 [0.643, 0.797]

[a] Regularised boosting (depth 2); other boosting rows use depth 3 and overfit more (training AUC 0.96–1.00).

All gliomas (314 scans, 157 patients) show the same pattern: clinical + timing with boosting reaches 0.777; adding radiomics gives 0.777, volumes 0.760 and all imaging 0.752; imaging-only models reach at most 0.695.

4.2 Does radiomics add to the non-imaging baseline?

Table 3 gives the paired comparison against the clinical + timing + treatment baseline.

Table 3: Radiomics added to the clinical + timing + treatment baseline (ridge). ΔAUC is paired against the baseline.

	Grade 4		All gliomas	
Model	AUC	ΔAUC	AUC	ΔAUC
Baseline	0.694	–	0.719	–
Best of 70 configurations (post hoc)	0.720	+0.024 [-0.032, +0.079]	0.754	+0.034 [-0.018, +0.082]
Nested choice over 70 (honest)	0.687	−0.008 [-0.046, +0.026]	0.733	+0.013 [-0.031, +0.054]
Configurations with ΔAUC CI > 0	0 / 70		0 / 70	

Correcting the radiomics extraction did not help. The small signal in the PRISM report features comes from texture computed on two grey levels, which acts as a description of the spatial pattern of bright and dark regions within the lesion; the corrected 32-bin features carried no additional signal.

In nested selection over feature groups, clinical + timing was part of the chosen combination in 22 of 25 outer folds (grade 4) and 24 of 25 (all gliomas). An imaging group was usually added

as well, but a different one from fold to fold, which is consistent with imaging contributing noise-level rather than stable signal.

4.3 PRISM report split

On the 30 test patients of the PRISM report split, with configurations fixed beforehand from cross-validation (Table 4), a model without imaging outperformed the PRISM report radiomics model, and adding radiomics changed it by at most +0.005. The same models average 0.72–0.74 under repeated cross-validation, so this split is favourable to every model.

Table 4: PRISM report split (127 training / 30 test patients, all gliomas).

Model	AUC (original label)	IPCW AUC (120 d)
PRISM report radiomics model	0.802	0.811
Clinical + timing + treatment, ridge	0.870	0.844
Clinical + timing + treatment, boosting	**0.901**	**0.881**
+ PRISM report radiomics (top 10)	0.869	0.848
+ corrected compartment radiomics (stacked)	0.869	0.843

4.4 External test on LUMIERE

Models trained on all MU grade-4 scans were applied unchanged to LUMIERE (Table 5). The PRISM report radiomics model, re-extracted with its original pipeline on 118 of the 132 scans, fell to 0.613. Among shared-feature models, volumes helped the boosting model (0.663 to 0.719) but not the ridge model (0.628 to 0.629); the confidence intervals overlap. Recalibrating the intercept and slope on other LUMIERE patients brought the observed/expected ratio to 0.94–0.98 without changing the ranking.

Table 5: Locked models on LUMIERE (132 scans, 61 patients), pooled IPCW AUC at 120 days.

Features	Learner	MU CV	LUMIERE [95% CI]
PRISM report radiomics model	–	–	0.613 [0.500, 0.729]
Clinical	ridge	0.578	0.609 [0.479, 0.735]
Clinical + timing	ridge	0.644	0.628 [0.521, 0.748]
Clinical + timing	boosting	0.714	0.663 [0.513, 0.781]
Volumes	ridge	0.591	0.534 [0.427, 0.629]
Clinical + timing + volumes	ridge	0.695	0.629 [0.505, 0.753]
Clinical + timing + volumes	boosting	0.709	**0.719** [0.609, 0.832]

5 Limitations

- **Treatment features are not yet restricted to what was known at the scan date.** Radiotherapy dose, duration and chemotherapy are used even for scans taken before radiotherapy ended (29 of 215 such scans). A one-off check with as-of-scan masking gave a *higher* baseline (grade 4 ridge 0.694 to 0.732), so the conclusion is not driven by this, but the tables will be re-run once the masking is in the code.
- **The label is partly shaped by the scan schedule.** Progression is usually declared at an MRI: 61 of 106 progressing patients have their progression date within three days of an MRI, and the interval to the next MRI alone (not available at prediction time) gives AUC 0.64 on grade 4. Part of the timing signal may reflect closer follow-up early after radiotherapy rather than tumour biology.

- **Scope.** Conventional MRI only (no perfusion or diffusion), a single development centre, about 130–160 patients, and a 120-day horizon. The result does not show that imaging is uninformative about progression in general.
- **Multiple comparisons.** Many configurations were evaluated on the same folds. Post-hoc rows are optimistic; the nested and paired estimates are the ones to quote. The LUMIERE result of 0.719 is the best of 16 combinations.
- **External cohort differences.** LUMIERE defines progression by expert RANO rating on imaging, uses automatic segmentation in atlas space (its necrosis compartment likely includes the resection cavity), and lacks treatment fields.

6 Conclusion

A model that uses no imaging — clinical, timing and treatment information — performs as well as or better than models that also use the MRI, and imaging alone is weaker. No imaging addition reached statistical significance on MU-Glioma-Post. The supported claim is that, in this setting, *imaging does not add measurable value over clinical and treatment information*; it is not that clinical information beats imaging in general. Image-based progression models should be reported against a treatment-aware non-imaging baseline under patient-grouped, censoring-aware evaluation.

Data and code

Pipeline: `prism/` in this repository. Source results: the C2, C3, external LUMIERE and imaging-versus-non-imaging write-ups in `docs/results/`.

References

[1] MU-Glioma-Post. The Cancer Imaging Archive, `https://doi.org/10.7937/7K9K-3C83`; data descriptor Sci Data, `https://doi.org/10.1038/s41597-025-06011-7`.

[2] Suter Y, et al. The LUMIERE dataset: longitudinal glioblastoma MRI with expert RANO evaluation. Sci Data 2022, `https://doi.org/10.1038/s41597-022-01881-7`.